Chapter 3: Supply, Demand, and Market Equilibrium

Principles of Microeconomics Curriculum

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Abstract

This chapter provides a rigorous, pedagogical treatment of Chapter 3: Supply, Demand, and Market Equilibrium, benchmarked to standard principles curriculum.

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1 The Nature of Demand and Consumer Valuation

The Law of Demand states that, *ceteris paribus*, there is an inverse relationship between the price of a good and the quantity demanded:

$$P \uparrow \implies Q_d \downarrow, \qquad P \downarrow \implies Q_d \uparrow \tag{1}$$

This empirical regularity is driven by two microeconomic mechanisms:

1. **The Substitution Effect:** As the price of a good rises, it becomes relatively more expensive compared to alternative goods, incentivizing consumers to substitute toward cheaper alternatives.
2. **The Income Effect:** A price increase erodes the consumer's real purchasing power, constraining the total quantity of goods they can afford to purchase.

1.1 Direct vs. Inverse Demand

In empirical economics, demand is specified directly as quantity as a function of price:

$$Q_d = a - bP \quad (a > 0, b > 0) \tag{2}$$

Because economists traditionally graph price on the vertical axis and quantity on the horizontal axis (following Alfred Marshall), we invert the equation to obtain the **Inverse Demand Function**:

$$P = \frac{a}{b} - \frac{1}{b}Q_d \tag{3}$$

Here, $\frac{a}{b}$ represents the **choke price**—the price at which quantity demanded drops to exactly zero.

2 Shifts in Demand vs. Movements Along the Demand Curve

Crucial Distinction: Demand vs. Quantity Demanded

- **Movement Along the Curve (Change in Quantity Demanded):** Caused strictly by a change in the good's *own price*.
- **Shift of the Entire Curve (Change in Demand):** Caused by changes in non-price determinants, altering quantity demanded at every price level.

2.1 Non-Price Determinants of Demand

- **Consumer Income:** For a *normal good*, demand shifts rightward as income expands ($\varepsilon_I > 0$). For an *inferior good* (such as instant ramen or intercity bus transit), demand shifts leftward as income rises ($\varepsilon_I < 0$).
- **Prices of Related Goods:**
  - *Substitutes:* An increase in the price of Good Y shifts demand for Good X to the right ($\varepsilon_{xy} > 0$).

  - *Complements:* An increase in the price of Good Y shifts demand for Good X to the left ($\varepsilon_{xy} < 0$).

- **Tastes and Preferences:** Exogenous shifts in consumer culture, health studies, or advertising.
- **Buyer Expectations:** Anticipated future price increases prompt consumers to accelerate current purchases, shifting current demand rightward.
- **Market Demographics:** An expansion in the population of potential buyers shifts market demand outward.

3 The Nature of Supply and Marginal Opportunity Costs

The Law of Supply establishes a positive relationship between price and quantity supplied, *ceteris paribus*:

$$P \uparrow \implies Q_s \uparrow, \qquad P \downarrow \implies Q_s \downarrow \tag{4}$$

Supply slopes upward because production is governed by the **Law of Diminishing Marginal Productivity**. As firms hire variable factor inputs into fixed plant capacity, the marginal product of labor declines, forcing marginal production costs (MC) to escalate. To cover higher incremental costs, firms require higher market prices.

3.1 Linear Supply Formulations

$$Q_s = c + dP \iff P = -\frac{c}{d} + \frac{1}{d}Q_s \quad (d > 0) \tag{5}$$

3.2 Non-Price Determinants of Supply

Supply shifts when underlying production parameters change:

- **Input Prices:** Wage increases, raw material inflation, or soaring energy costs increase marginal cost, shifting supply leftward.
- **Technological Progress:** Cost-reducing manufacturing breakthroughs allow firms to produce more output at each price level, shifting supply rightward.
- **Government Taxes and Subsidies:** An excise tax increases per-unit production cost, shifting supply vertically upward by the tax amount t.
- **Number of Active Producers:** Industry entry expands market supply; business closures contract it.

4 Market Equilibrium and Price Discovery

A competitive market attains equilibrium where quantity demanded equals quantity supplied:

$$Q_d(P^*) = Q_s(P^*) = Q^* \tag{6}$$

Step-by-Step Algebraic Equilibrium Solution

Consider a regional agricultural market for specialty roasted coffee beans:

$$Q_d = 500 - 15P, \qquad Q_s = -100 + 15P$$

1. Equate Q_d and Q_s:

$$500 - 15P = -100 + 15P \implies 30P = 600 \implies P^* = \$20$$

2. Substitute $P^* = \$20$ into either direct schedule:

$$Q^* = 500 - 15(20) = 500 - 300 = 200 \text{ bags}$$

3. Verify at disequilibrium prices:

- If $P = \$25$ (Price too high): $Q_d = 125$, $Q_s = 275 \implies$ Surplus $= +150$. Unsold inventory forces price downward.
- If $P = \$15$ (Price too low): $Q_d = 275$, $Q_s = 125 \implies$ Shortage $= -150$. Eager buyers bid price upward.

5 Simultaneous Dual Shocks and the Indeterminacy Matrix

When demand and supply shift concurrently, exactly one equilibrium variable (P^* or Q^*) changes unambiguously, while the other is indeterminate without knowing the exact relative magnitudes of the shifts.

Simultaneous Shock	**Equilibrium Price (P^*)**	**Equilibrium Quantity (Q^*)**
Demand ↑, Supply ↑	Ambiguous	Unambiguously Rises (↑)
Demand ↓, Supply ↓	Ambiguous	Unambiguously Falls (↓)
Demand ↑, Supply ↓	Unambiguously Rises (↑)	Ambiguous
Demand ↓, Supply ↑	Unambiguously Falls (↓)	Ambiguous